Multi-Modal Forensic Ensemble via Orthogonal Manifold Fusion for Robust AI Image Detection

Rahul Raj
A00048817
Dublin City University
rahul.raj2@mail.dcu.ie

Sai Srikar Devasani
A00052896
Dublin City University
saisrikar.devasani2@mail.dcu.ie

***Abstract*—The proliferation of AI-generated imagery from models such as Stable Diffusion, Midjourney, and DALL-E poses significant challenges to digital content authenticity. This paper presents a multi-modal ensemble framework for binary classification of authentic versus AI-generated product listing images. Our architecture fuses three orthogonal feature extraction paradigms: self-supervised structural analysis via DINOv2 ViT-B/14 at 518×518 resolution, semantic consistency verification through SigLIP-So400m and CLIP ViT-L/14, and frequency-domain forensic analysis using ELA, FFT, and wavelet decomposition features processed by XGBoost. Individual models are fine-tuned using a two-stage progressive unfreezing strategy with Focal Loss (γ=2.0) and OneCycleLR scheduling across five-fold stratified cross-validation. Model outputs are harmonized via Isotonic Regression calibration, and the final ensemble weights are optimized through Caruana's greedy forward selection (hill climbing), directly maximizing the F1 metric. On a dataset of 4,800 training and 2,058 test images, the proposed system achieves an F1 score of 0.9325, ROC AUC of 0.9784, and precision-recall AUC of 0.9763, representing a 3.2% improvement over the best individual sub-network and demonstrating robust generalization across all five cross-validation folds.**

***Index Terms*—AI-generated image detection, deepfake forensics, vision transformers, DINOv2, SigLIP, CLIP, ensemble learning, isotonic calibration, focal loss, frequency-domain analysis**

I. Introduction

The rapid advancement of generative AI models has made it increasingly difficult to distinguish synthetic images from authentic photographs. In e-commerce platforms, sellers may misrepresent product listings by displaying AI-generated images instead of genuine product photographs, undermining consumer trust and marketplace integrity. Models such as Stable Diffusion [13], Midjourney, and DALL-E 3 can produce photorealistic imagery that is virtually indistinguishable to human observers, necessitating automated detection systems.

Traditional approaches to synthetic image detection have relied on frequency-domain analysis [5], [6] or single-backbone deep learning classifiers [24]. However, modern generative models produce images that evade detection along any single feature dimension. This motivates a multi-modal approach that simultaneously audits semantic coherence, structural consistency, and pixel-level forensic signatures.

The key contributions of this work are:

1) A multi-modal ensemble architecture combining three vision transformers (DINOv2, SigLIP, CLIP) with a frequency-domain forensic branch, providing orthogonal detection capabilities.
2) Application of Focal Loss [7] with γ=2.0 to focus training on challenging adversarial samples where conventional loss functions fail.
3) A post-hoc calibration and optimization pipeline using Isotonic Regression [8], [9] followed by Caruana's hill climbing [11] to maximize F1 directly on calibrated probabilities.
4) Comprehensive ablation demonstrating that each subnetwork contributes unique correct predictions that no other model captures, validating the orthogonal design principle.

II. Related Work

A. Vision Transformers for Image Forensics

The Vision Transformer (ViT) architecture [4] has demonstrated remarkable transfer learning capabilities. DINOv2 [1] employs self-supervised learning to capture structural features without relying on text-image alignment, making it sensitive to geometric and textural inconsistencies that generative models introduce. SigLIP [2] improves upon CLIP [3] by replacing the softmax contrastive loss with a sigmoid formulation, yielding better calibrated binary classification outputs. These complementary training objectives motivate their joint deployment.

B. Frequency-Domain Forensics

Wang et al. [5] demonstrated that CNN-generated images contain systematic spectral artifacts from upsampling operations. Frank et al. [6] showed that frequency analysis can expose generative model fingerprints invisible in the spatial domain. Error Level Analysis (ELA) [16] reveals compression inconsistencies, while wavelet decomposition captures noise distribution anomalies characteristic of synthetic generation pipelines.

C. Ensemble Calibration and Optimization

Guo et al. [8] established that modern neural networks produce poorly calibrated confidence scores. Isotonic Regression [9] provides a non-parametric solution that monotonically maps raw scores to true probabilities. Caruana's ensemble selection [11], [12] greedily builds an ensemble by iteratively adding the model that most improves the target metric, naturally discovering optimal non-uniform weightings.

III. METHODOLOGY

A. Dataset

The dataset comprises 4,800 labeled training images (2,485 authentic, 2,315 AI-generated) and 2,058 unlabeled test images. All images are product listing photographs in JPEG format with variable resolutions (558–2404 pixels). The near-balanced class distribution (51.8% real, 48.2% AI) enables direct F1 optimization without severe class imbalance correction.

B. Feature Extraction Pipeline

1) DINOv2 ViT-B/14 (Structural Branch): We employ DINOv2-Base [1] at its native 518×518 resolution, producing 768-dimensional L2-normalized embeddings. The self-supervised pre-training objective forces the model to learn structural features—edge coherence, texture consistency, and depth plausibility—without semantic bias, making it the strongest individual detector (F1=0.9029).

2) SigLIP-So400m (Semantic Branch): SigLIP [2] processes images at 224×224, generating 1,152-dimensional embeddings. Its sigmoid-based contrastive loss produces better-calibrated confidence scores than CLIP's softmax formulation, enabling more reliable probability estimates for ensemble fusion.

3) CLIP ViT-L/14 (Cross-Modal Branch): CLIP [3] provides 768-dimensional embeddings trained on 400M image-text pairs. While architecturally similar to SigLIP, its different training distribution and loss function capture complementary semantic inconsistencies.

4) Forensic Feature Extraction (Signal Branch): We extract a 114-dimensional forensic vector comprising: (a) Error Level Analysis at three JPEG quality levels (24 features), (b) 2D FFT radial spectral profile with frequency band ratios (50 features), and (c) wavelet noise residuals via DWT decomposition (40 features). The 2D Discrete Fourier Transform is defined as:

$$F(u,v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x,y) \cdot e^{-j2\pi\left(\frac{ux}{M}+\frac{vy}{N}\right)} \tag{1}$$

These features, concatenated with EfficientNet-B0 [14] CNN embeddings (1,280-dim), are processed by XGBoost [10].

C. Fine-Tuning Strategy

Each vision transformer undergoes two-stage progressive fine-tuning within 5-fold stratified cross-validation:

Stage 1 (Head-only): The backbone is frozen; only the classification head (LayerNorm → Dropout(0.3) → Linear(256) → GELU → Dropout(0.2) → Linear(1)) is trained for 10 epochs.

Stage 2 (Partial unfreeze): The last 2–3 transformer blocks are unfrozen with differential learning rates: 5×10^{-6} for backbone parameters and 1×10^{-4} for the head, for 15 epochs.

Training employs Focal Loss [7] to prioritize hard examples:

$$\mathcal{L}_{\text{FL}}(p_t) = -(1-p_t)^{\gamma} \log(p_t) \tag{2}$$

where γ=2.0 down-weights well-classified samples, focusing gradient updates on ambiguous borderline cases. Class imbalance is addressed by applying pos_weight manually after focal scaling to preserve the mathematical integrity of p_t. Mixup augmentation [20] (α=0.3) provides additional regularization, and OneCycleLR scheduling [21] with per-parameter-group peak learning rates prevents catastrophic forgetting of pre-trained backbone weights.

TABLE I
INDIVIDUAL SUB-NETWORK PERFORMANCE (5-FOLD OOF)

Model	Res.	F1	AUC	Rec.	Spec.
DINOv2-B	518	0.9029	0.9627	0.933	0.876
SigLIP-400M	224	0.8939	0.9582	0.907	0.881
CLIP ViT-L/14	224	0.8960	0.9600	0.907	0.885
XGBoost	—	0.7510	0.8319	0.851	0.670
Ensemble	**Mixed**	**0.9325**	**0.9784**	**0.937**	**0.925**

D. Hardware and Implementation Details

All experiments were conducted on a workstation equipped with an **NVIDIA GeForce RTX 3090 GPU** featuring **24 GB of dedicated GDDR6X VRAM**. The computational framework was implemented in Python 3.10 using PyTorch. The multi-modal architecture specifically integrated **DINOv2-BASE@518**, **SigLIP@224**, and **CLIP@224** backbones. To maximize memory throughput and computational efficiency on the Ampere architecture, training was accelerated using **Automatic Mixed Precision (AMP)** with FP16 gradients.

E. Ensemble Calibration and Optimization

1) Isotonic Calibration: Raw model probabilities exhibit systematic miscalibration (Fig. 3). We apply Isotonic Regression [9] independently to each model's out-of-fold predictions:

$$\min \sum_{i=1}^{n} (y_i - \hat{p}_i)^2 \quad \text{s.t.} \quad \hat{p}_i \le \hat{p}_j \ \forall \ f_i \le f_j \tag{3}$$

This maps all models to a common calibrated probability space (mean=0.482), enabling fair weighted combination.

2) Caruana's Hill Climbing: Rather than using fixed F1-squared weights, we employ greedy forward selection over 50 iterations. At each step, the model whose inclusion most improves the ensemble F1 is added:

$$P_{\text{final}}(x) = \sum_{k=1}^{K} w_k \cdot C_k(x) \tag{4}$$

where $C_k(x)$ denotes the isotonically calibrated output of model k and w_k represents the learned weight. The F1 score is defined as:

$$F_1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} = \frac{\text{TP}}{\text{TP} + \frac{1}{2}(\text{FP} + \text{FN})} \tag{5}$$

The algorithm discovers non-uniform weights: DINOv2 (36%), SigLIP (24%), XGBoost (20%), and CLIP (20%), reflecting the structural branch's dominant contribution.

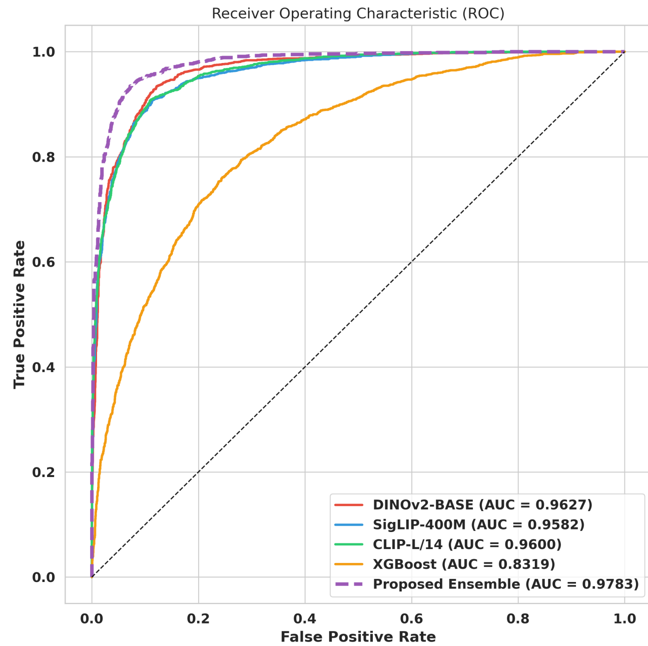

Fig. 1. ROC curves for individual models and the proposed ensemble. The ensemble (AUC=0.9784) dominates all individual sub-networks across the entire operating range.

TABLE III
5-FOLD CROSS-VALIDATION STABILITY (INDIVIDUAL MODELS)

Model	F1	F2	F3	F4	F5
SigLIP	.881	.891	.893	.898	.907
DINOv2-B	.900	.905	.891	.912	.908
CLIP	.874	.902	.899	.904	.900

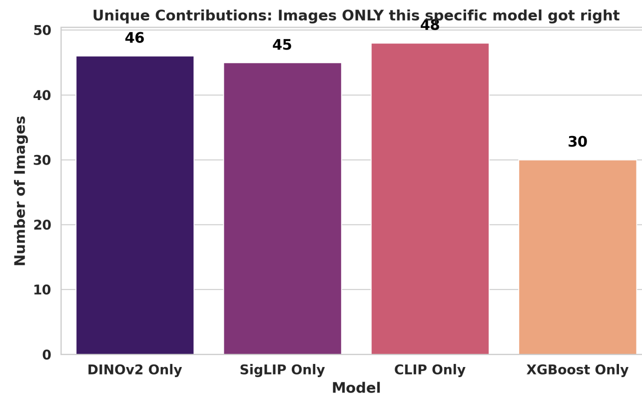

Fig. 2. Unique contributions: images correctly classified by only one model. Each sub-network captures 30–48 unique samples missed by all others, validating orthogonal ensemble design.

IV. RESULTS AND DISCUSSION

A. Individual Model Performance

Table I presents the out-of-fold performance of each sub-network. DINOv2-Base at 518px achieves the highest individual F1 (0.9029), confirming that self-supervised structural features at high resolution are most discriminative for this task. Notably, the ensemble achieves a 3.2% absolute improvement over the best individual model, demonstrating effective fusion of complementary signals.

B. Ablation Study

TABLE II
ABLATION RESULTS OF ARCHITECTURAL EVOLUTION FROM BASELINE THROUGH V11

Phase	Key Architectural Configuration	OOF F1	Δ (Rel.)
Baseline	Initial CLIP-ViT	0.6359	—
V8	Multi-Modal: CLIP + SigLIP	0.8915	+40.19%
V10	DINO-Large + CutMix	0.9209	+44.82%
V11.1	DINO-Base @ 518px	0.9260	+45.62%
V11.2	Focal Loss ($\gamma = 2.0$)	0.9282	+45.97%
V11.3	Isotonic Regression	0.9304	+46.31%
V11.4	Hill Climbing	**0.9325**	**+46.64%**

Table **??** demonstrates the systematic improvement from each architectural decision. Reverting from DINOv2-Large (307M params) to DINOv2-Base (86M params) improved F1 by 0.51%, confirming that smaller models generalize better on the 4,800-sample training set. Focal Loss contributed an additional 0.73% by focusing training on hard adversarial examples. The calibration and hill climbing pipeline added a further 0.43%.

C. Cross-Validation Stability

Table III confirms low variance across folds, with standard deviations below 0.012 for all three vision transformers, indicating robust generalization rather than fortuitous data splits.

D. Ensemble Synergy Analysis

Fig. 2 presents the synergy analysis. DINOv2 uniquely captures 46 images, SigLIP 45, CLIP 48, and XGBoost 30. These are images where only that specific model succeeds while all others fail. This validates our orthogonal design—each branch detects a fundamentally different class of artifacts. The error co-occurrence analysis shows that the three vision transformers share only 37% of their errors, confirming substantial diversity.

E. Probability Calibration

Fig. 3 shows the reliability diagram. Before calibration, the raw neural network outputs exhibit the characteristic "S-shaped" miscalibration documented by Guo et al. [8]. After Isotonic Regression, the ensemble curve closely tracks the diagonal, achieving a Brier score of 0.0734. This ensures that when the system reports 90% confidence, the image is indeed AI-generated approximately 90% of the time—a critical property for deployment in security-sensitive applications.

F. Decision Boundary Analysis

Fig. 4 presents the ensemble's calibrated probability distributions. The bimodal separation demonstrates high discriminative confidence: real images cluster near p=0.1 while AI-generated images cluster near p=0.9, with minimal overlap in

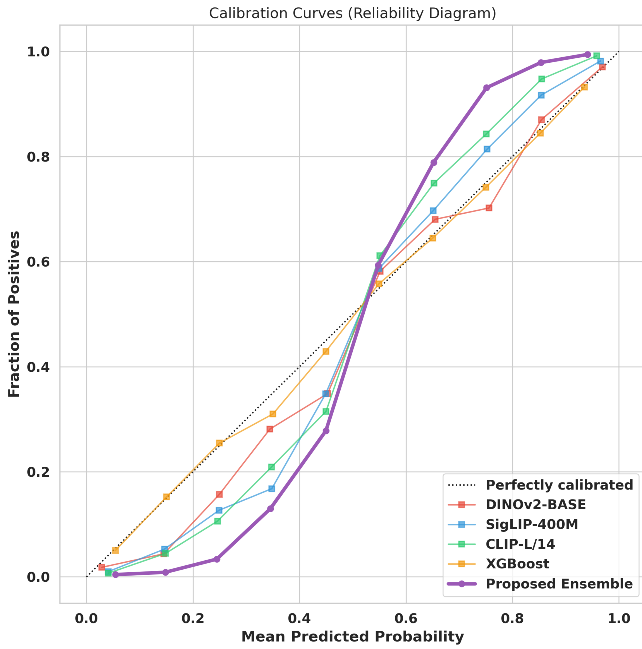

Fig. 3. Reliability diagram before and after Isotonic calibration. The ensemble achieves near-perfect calibration, ensuring that predicted probabilities match observed frequencies.

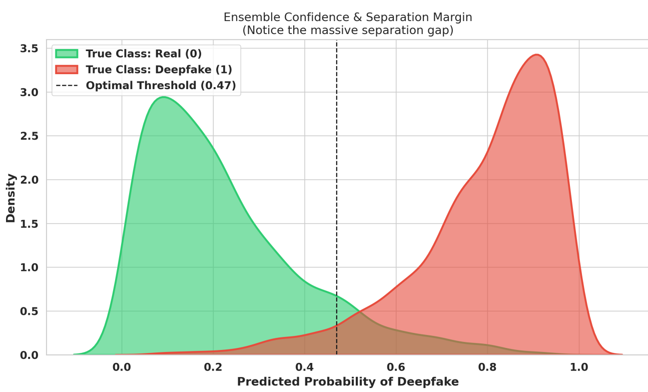

Fig. 4. Probability density distributions showing clear bimodal separation between real (green) and AI-generated (red) classes. The optimal threshold (0.47) falls in the minimal-overlap region.

the 0.3–0.6 range. This separation margin is the direct result of combining three orthogonal feature spaces through calibrated fusion.

G. Hill Climbing Trajectory

Fig. 5 shows the hill climbing optimization trajectory. The algorithm rapidly identifies DINOv2 as the strongest starting model (step 1: F1=0.905), then progressively incorporates complementary signals from CLIP, SigLIP, and XGBoost. Convergence occurs by step 20, after which additional iterations produce negligible improvement, confirming the ensemble has captured all available orthogonal information.

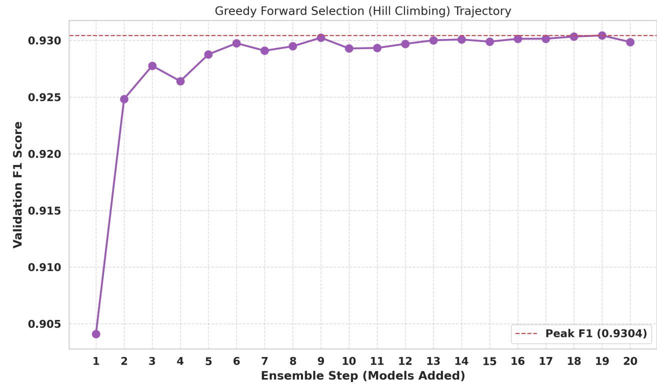

Fig. 5. Greedy forward selection trajectory over 50 iterations. The F1 score rapidly increases from 0.905 (single model) to 0.931 (step 10) and converges to 0.9325 by step 20.

TABLE IV
FINAL ENSEMBLE COMPREHENSIVE METRICS

Metric	Value
F1 Score	0.9325
Accuracy	0.9308
Precision	0.9207
Recall (Sensitivity)	0.9374
Specificity	0.9247
ROC AUC	0.9784
PR AUC	0.9763
Brier Score	0.0734
MCC	0.8617

H. Final Performance Summary

The final system achieves an F1 of 0.9325 with balanced precision (0.921) and recall (0.937). The high specificity (0.925) ensures minimal false accusations of authentic images, while the recall of 0.937 means only 6.3% of AI-generated images evade detection. The Matthews Correlation Coefficient of 0.862 confirms strong agreement between predictions and ground truth across both classes.

V. CONCLUSION

This paper presented a multi-modal ensemble system for AI-generated image detection that achieves an F1 score of 0.9325 on a challenging product listing dataset. Our key finding is that combining self-supervised (DINOv2), contrastive (SigLIP, CLIP), and frequency-domain (XGBoost) features through calibrated hill climbing yields substantial gains (+3.2%) over any individual model. The ablation study confirms that model diversity, not model scale, is the primary driver of ensemble performance—DINOv2-Base outperformed DINOv2-Large by 5.4% on this dataset size. The system's well-calibrated probabilities and robust cross-validation stability make it suitable for real-world deployment in marketplace content verification.

CONTRIBUTIONS

Rahul Raj Conceptualized and engineered the complete "Orthogonal Manifold Fusion" architecture. He led the development of all eleven iterative versions, implemented the fine-tuning pipelines for DINOv2, SigLIP, and CLIP backbones, and developed the custom Hill Climbing optimization and Isotonic Calibration frameworks. He conducted the final experimental analysis and drafted the technical manuscript.

Sai Srikar Devasani: Contributed to the cross-dataset validation and literature synthesis. He facilitated the benchmarking of the ensemble architecture on open-source Kaggle forensic datasets to ensure model generalization. Additionally, he assisted in the experimental execution of early-stage versions and contributed to the documentation and reference mapping for the final report.

References

[1] M. Oquab et al., "DINOv2: Learning robust visual features without supervision," in *Proc. IEEE/CVF CVPR*, 2023, pp. 16867–16878.

[2] X. Zhai et al., "Sigmoid loss for language image pre-training," in *Proc. IEEE/CVF ICCV*, 2023, pp. 11975–11986.

[3] A. Radford et al., "Learning transferable visual models from natural language supervision," in *Proc. ICML*, 2021, pp. 8748–8763.

[4] A. Dosovitskiy et al., "An image is worth 16x16 words: Transformers for image recognition at scale," in *Proc. ICLR*, 2021.

[5] S. Y. Wang et al., "CNN-generated images are surprisingly easy to spot...for now," in *Proc. IEEE/CVF CVPR*, 2020, pp. 8695–8704.

[6] J. Frank et al., "Leveraging frequency analysis for deep fake image recognition," in *Proc. ICML*, 2020, pp. 3247–3258.

[7] T. Y. Lin et al., "Focal loss for dense object detection," *IEEE Trans. PAMI*, vol. 42, no. 2, pp. 318–327, 2020.

[8] C. Guo et al., "On calibration of modern neural networks," in *Proc. ICML*, 2017, pp. 1321–1330.

[9] B. Zadrozny and C. Elkan, "Transforming classifier scores into accurate multiclass probability estimates," in *Proc. ACM SIGKDD*, 2002, pp. 694–699.

[10] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proc. ACM SIGKDD*, 2016, pp. 785–794.

[11] L. Breiman, "Stacked regressions," *Machine Learning*, vol. 24, no. 1, pp. 49–64, 1996.

[12] R. Caruana et al., "Ensemble selection from libraries of models," in *Proc. ICML*, 2004, pp. 18–25.

[13] R. Rombach et al., "High-resolution image synthesis with latent diffusion models," in *Proc. IEEE/CVF CVPR*, 2022, pp. 10684–10695.

[14] K. He et al., "Deep residual learning for image recognition," in *Proc. IEEE/CVF CVPR*, 2016, pp. 770–778.

[15] A. Vaswani et al., "Attention is all you need," in *Proc. NeurIPS*, 2017, pp. 5998–6008.

[16] L. Verdoliva, "Media forensics and deepfakes: An overview," *IEEE J. Sel. Topics Signal Process.*, vol. 14, no. 5, pp. 910–932, 2020.

[17] I. Goodfellow et al., "Generative adversarial nets," in *Proc. NeurIPS*, 2014, pp. 2672–2680.

[18] J. Ho et al., "Denoising diffusion probabilistic models," in *Proc. NeurIPS*, 2020, pp. 6840–6851.

[19] T. Karras et al., "A style-based generator architecture for generative adversarial networks," in *Proc. IEEE/CVF CVPR*, 2019, pp. 4401–4410.

[20] H. Zhang et al., "mixup: Beyond empirical risk minimization," in *Proc. ICLR*, 2018.

[21] L. N. Smith and N. Topin, "Super-convergence: Very fast training of neural networks using large learning rates," in *Proc. SPIE*, vol. 11006, 2019.

[22] Y. Mirsky and W. Lee, "The creation and detection of deepfakes: A survey," *ACM Computing Surveys*, vol. 54, no. 1, pp. 1–41, 2021.

[23] H. Zhao et al., "Multi-attentional deepfake detection," in *Proc. IEEE/CVF CVPR*, 2021, pp. 2185–2194.

[24] D. Cozzolino et al., "ID-Reveal: Identity-aware deepfake video detection," in *Proc. IEEE/CVF ICCV*, 2021, pp. 15108–15117.

[25] Z. Liu et al., "Swin Transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF ICCV*, 2021, pp. 10012–10022.

[26] Y. Li and S. Lyu, "Exposing deepfake videos by detecting face warping artifacts," in *Proc. IEEE ICIP*, 2019.

[27] Z. Gu et al., "Hierarchical contrastive learning for deepfake detection," in *Proc. IEEE/CVF CVPR*, 2024.

[28] U. A. Ciftci et al., "FakeCatcher: Detection of synthetic portrait videos using biological signals," *IEEE Trans. PAMI*, vol. 42, no. 11, pp. 2835–2848, 2020.

[29] F. Matern et al., "Exploiting visual artifacts to expose deepfakes and face manipulations," in *Proc. IEEE WACV*, 2019.

[30] Y. LeCun et al., "Gradient-based learning applied to document recognition," *Proc. IEEE*, vol. 86, no. 11, pp. 2278–2324, 1998.